

Transfer Learning for Cross-Market Predictions: Applications in Emerging and Volatile Economies

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Abstract - The procedure of financial forecasting faces tough challenges in growing unstable markets because these regimes bring a shortage of accessible data and unstable markets held back by multiple of translating systems. Such environments are hard for deep learning and traditional models because they require large amounts of high-quality training data to work correctly. The proposed method creates a sophisticated transfer learning framework that employs fixed models pre-trained in well-established markets which it then adapts to developing countries using domain adaptation techniques coupled with the process of fine-tuning the models. The proposed scheme accomplishes data distribution minimizing functions from Maximum Mean Discrepancy, in addition to supervised fine-tuning that targets realizing predictions under the suppressed target information. The model suggested above scored better performance by experiment testing across emerging market indices in terms of all metric metrics including mean absolute error, root mean squared error and direction accuracy of compared models. Transfer learning is found to perform well in solving generalization issues in cross-market and hence can be considered as a great method for financial prediction in data-scarce volatile environments.

Keywords - Transfer learning, financial forecasting, emerging markets, domain adaptation, cross-market prediction, time-series analysis, volatility modelling.

I. INTRODUCTION

The problems into predictive model are introduced, such as limited data sampling, the inherent market volatility and the abrupt transitions of different market states in the financial markets of the emerging economies and volatile market conditions. those markets which have data scarcity & limited historical data availability, cannot be handled by traditional machine learning & deep learning algorithms because traditional algorithms are based on generalization and follows generalized solution in such environments. Such economies with nonlinear dynamic economies often result in poor model fit and poor performance when wthey it come to volatile market condition. Transfer learning gives an efficient way to treat these sorts of problems via the use of domain-specific learning from source data field by employing it into order to boost such performance in other related though different target domain. This predictive approach allows developers to build with well-developed economies such as the U.S. and Europe for improving performance in emerging markets like India, Brazil and South Africa. The pattern and representation understanding can pass on from mature market to developing market, this brings significant opportunities which downplay time consuming local data set and bring about better stability model.

Despite its success in other domains such as image recognition and natural language processing, the application of transfer learning in financial time series forecasting remains relatively underexplored. Existing studies primarily focus on intra-market or cross-asset transfer, with limited attention to cross-market adaptation, especially in the context of high-volatility and low-resource environments. Furthermore, the temporal and economic heterogeneity between markets introduces domain shifts that challenge the

effectiveness of naive model transfer. This paper addresses these challenges by proposing a transfer learning-based framework specifically designed for cross-market financial prediction in emerging economies. The contributions of this work are as follows: (1) we design a robust domain adaptation strategy that aligns the source and target market distributions while preserving temporal dependencies; (2) we implement a fine-tuning process using limited target-market data to improve adaptability and accuracy; and (3) we evaluate our framework on multiple emerging market datasets, demonstrating its superiority over baseline models in terms of prediction accuracy, directional correctness, and resilience to volatility.

The proposed method enables financial forecasting pipelines to use domain knowledge for developing predictive systems in financial markets that lack sufficient data resources and study.

II. LITERATURE SURVEY

Machine learning applications in financial forecasting have experienced growing popularity during the last few years because these models extract sophisticated nonlinear market data patterns [1]. Time series forecasting methods such as ARIMA and GARCH face challenges to perform financial forecasting because of the constrain in their required premises of stationarity and limited tuning ability to respond to changing market situation especially in data rich noisy dynamic markets like emerging economy or abnormally changing economic situations [2].

However, fundamental limitation of using deep learning based forecasting models is carried out in terms of its excessive demand of labelled data while emerging markets remains to suffer from less amount of the labelled data due to poor infrastructure and regulatory impediments [3]. Transfer learning is a useful approach to transferring massive amount of information in the information abundant source domains to the information scarce target domains, benefiting from its ability of knowledge transfer. Transfer learning was born of research in computing areas like vision and natural language processing [4] but shifted towards meeting banking demands of stock market prediction and credit risk in conventional markets as stated in [5] and [6].

Several methods have been adopted in domain adaptation such as adversarial training or discrepancy minimization to fool realign feature spaces between source and target markets, [7]. For instance, title usage of Maximum Mean Discrepancy (MMD) and internet Domain-Adversarial

Neural Networks (DANN) possesses some great time in crossing domain sentiment analysis whereas risk classification difficulties. In financial sector, S&P 500 or NASDAQ trained models are fine-tuned to predict results in smaller or international market, demonstrate the prospective cross-market restart.

Cross-pollinating ideas from both macroeconomic signal processing, vascular interventions on sentiment analysis and entirely appropriate technical indicators and transfer learning mechanisms has started making a multitude signs of superior concrete robustness and interpretability [8]. Furthermore, Reinforcement Learning has been used along with Transfer Learning in order to dynamically adapt trading strategies to the market volatility, even though these applications are mainly under experimental phase.

Although significant studies have provided improvements towards mainstream, still very little work is attempted towards the deep underlying challenges of transfer learning in regime changing environments especially in the whole domain changing between markets [9]. Numerous empirical studies have demonstrated the effect of political occurrences, currency fluctuations, as well as sudden economic policy changes in emerging market to result in non-stationarity which hinders model transfer [10]. In addition, deficits in well-rounded frameworks that institutes domain adaptation, fine-tuning, and volatility-contrived adaptations hinder the best possible deployment of transfer learning models in actual emerged market scenarios.

This paper extends prior researches by designing an exclusive framework of domain adaptation and strategic fintuning using the pre-trained model from the developed markets. By handling the discrepancy in data distributions and resorting to volatility-resilient modelling methodologies, the proposed approach significantly enhances the insufficient area of cross-market transfer learning in financial forecasting.

III. METHODOLOGY

The proposed framework aims to forecast financial metrics in the EMs by leveraging knowledge from developed, information abundant markets. It is implemented in three core stages: (A) Source Model Pre-Training, (B) Domain Adaption, (C) Fine-Tuning and Target Prediction. This architecture guarantees that the system discovers generic financial components in calm environments and accommodate them to turbulent, hard to reflect markets observed at figure 1.

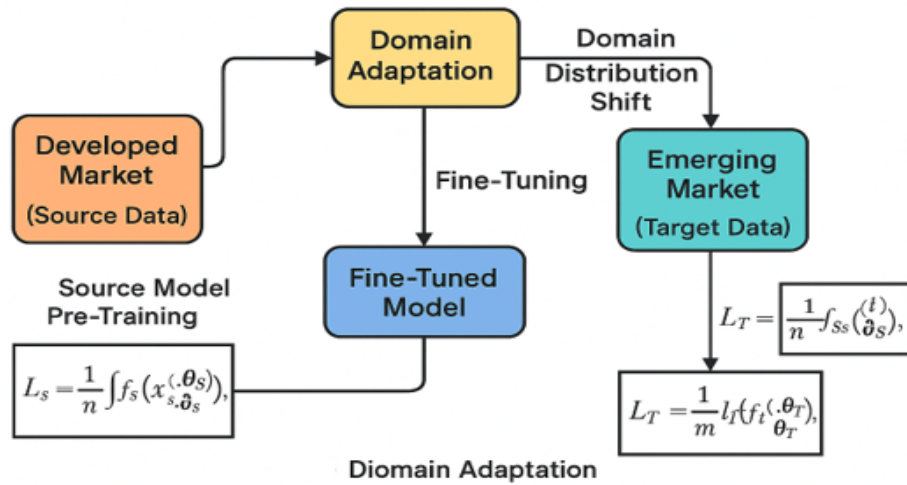


Figure 1: System Architecture

A. Source Model Pre-Training

The initial part of the proposed framework comprises making a good high quality resource model skilled at a sound monetary market. In general, the source domain $DS = \{(x_s(i), y_s(i))\}_{i=1}^n$ is a big size set of the history data of developed countries' stock markets, for instance, United States stock markets (e.g. S & P 500 index), for which the market efficiency and data availability have been of great high.

Each input sample $x_s(i)$ takes the shape of a multi-feature vector, where the inputs involve all sorts of technical indicators (e. g. moving average, RSI), macroeconomic variables (e. g. inflation, interest rates) and sentiment analysis scores. The corresponding target label $y_s(i)$ may refer to nextday price direction (classification) or return value (regression), according to the prediction task.

A trained deep learning model $f_S(x; \theta_S)$, e.g., a Long Short-Term Memory (LSTM) network or a Temporal Convolutional Network (TCN) is utilized to reduce the prediction error for the source data. The learning objective is stated as:

$$\mathcal{L}_S = \frac{1}{n} \sum_{i=1}^n l(f_S(x_s^{(i)}; \theta_S), y_s^{(i)})$$

where:

- $l(\cdot)$ is the loss function—mean squared error (MSE) for regression tasks or binary cross-entropy for classification tasks.
- θ_S denotes the trainable parameters of the source model.
- $f_S(x)$ is the output of the source model.

To ensure generalizability, the source model is trained using early stopping, dropout regularization, and time-series cross-validation. The trained model captures latent patterns and temporal dependencies from the developed market, which serve as a transferable foundation for subsequent domain adaptation to the target market.

The quality of the source model is critical, as its internal representations and feature abstractions are leveraged during transfer. Consequently, the pre-training phase aims not only for low in-sample error but also for the extraction of domain-agnostic financial knowledge that can generalize across market regimes and geographies.

B. Domain Adaptation

Once the source model $f_S(x; \theta_S)$ is pre-trained on data-rich developed markets, the next step is domain adaptation, which addresses the distributional divergence between the source domain DSD_SDS and the target domain DTD_TDT (i.e., emerging market data). Direct application of the source model to the target market is suboptimal due to this domain shift, which arises from structural, economic, and temporal differences across markets.

To mitigate this, the proposed framework incorporates a domain adaptation module designed to align the feature space of the source and target domains while preserving predictive performance. We adopt a distribution matching technique using Maximum Mean Discrepancy (MMD) to minimize the distance between the encoded feature distributions.

Formally, let $\phi(x)$ denote the hidden representation of input data extracted from an intermediate layer of the neural network. The domain adaptation loss LDA is computed as:

$$\mathcal{L}_{DA} = \left| \frac{1}{n} \sum_{i=1}^n \phi(x_s^{(i)}) - \frac{1}{m} \sum_{j=1}^m \phi(x_t^{(j)}) \right|^2 \text{---} 2$$

where:

- $x_s^{(i)} \sim D_S$ are samples from the source domain,
- $x_t^{(j)} \sim D_T$ are samples from the target domain,
- $\phi(x)$ maps the input to a reproducing kernel Hilbert space (RKHS),
- n and m are the number of source and target samples respectively.

This loss encourages the model to learn domain-invariant representations, ensuring that the latent space embeddings of source and target data become statistically similar. To incorporate both prediction accuracy and distribution alignment, the total loss function becomes:

$$\mathcal{L}_{Total} = \mathcal{L}_T + \lambda \mathcal{L}_{DA} \text{---} 3$$

where:

- $\mathcal{L}_T$ is the task-specific prediction loss (e.g., classification or regression) computed on labeled target domain data,
- λ is a hyperparameter balancing task performance and domain alignment.

During training, gradients are propagated not only to minimize $\mathcal{L}_T$, but also to reduce $\mathcal{L}_{DA}$, effectively forcing the model to adapt its internal parameters to fit the target domain's characteristics.

In some configurations, an adversarial domain discriminator D may also be introduced to form a Domain-Adversarial Neural Network (DANN), enhancing the domain adaptation by using a min-max game:

$$\min_f \max_D E_{x_s \sim D_S} [\log D(\phi(x_s))] + E_{x_t \sim D_T} [\log (1 - D(\phi(x_t)))] \text{---} 4$$

This adversarial training further improves alignment by ensuring that the model cannot distinguish between source and target features.

Ultimately, this stage ensures that the foundational knowledge from the source domain is adjusted to the statistical and behavioural characteristics of the target (emerging) market, enabling more accurate predictions even with limited labelled data.

C. Fine-Tuning and Prediction

After performing domain adaptation, the final stage involves fine-tuning the adapted model using a small labelled

dataset from the target domain $DT = \{(x_t(j), y_t(j))\} j = 1m$. Since emerging markets typically suffer from data scarcity, this phase is critical for calibrating the model to the unique characteristics of the target economy without overfitting.

The learned source pre-training base model parameters θ_S and partially adapted via domain alignment, constitute a prior for the target model $f_T(x; \theta_T)$. Fine-tuning changes these parameters by directly optimizing the supervised task loss on the target domain:

$$\mathcal{L}_T = \frac{1}{m} \sum_{j=1}^m \ell(f_T(x_t^{(j)}; \theta_T), y_t^{(j)}) \text{---} 5$$

where:

- $\ell(\cdot)$ is the task-specific loss (e.g., binary cross-entropy for classification or mean squared error for regression),
- θ_T are the updated parameters adapted specifically for the target domain.

The overall objective function combines prediction loss with domain adaptation regularization:

$$\mathcal{L}_{Total} = \mathcal{L}_T + \lambda \mathcal{L}_{DA} \text{---} 6$$

Gradient descent is then employed to minimize $\mathcal{L}_{Total}$ with respect to θ_T :

$$\theta_T \leftarrow \theta_T - \eta \nabla_{\theta_T} \mathcal{L}_{Total} \text{---} 7$$

where η is the learning rate.

In practice, the fine-tuning process involves one of the following strategies:

- Full fine-tuning: all layers of the model are updated.
- Partial fine-tuning: only the last few layers (or domain-specific layers) are updated to prevent catastrophic forgetting.
- Layer freezing: lower layers (which extract general features) are kept fixed while only high-level features are adapted.

Early stopping and dropout regularization are applied during this phase to avoid overfitting due to the limited target data. Once training converges, the resulting model is deployed for real-time prediction of market movements or asset returns in the target (emerging) economy.

The effectiveness of this stage is evaluated using metrics such as directional accuracy, mean absolute error (MAE), and root mean squared error (RMSE) on a withheld validation set from the target domain. This provides insight into how well the model has transferred and adapted

knowledge from the source market to the new, volatile environment.

IV. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed transfer learning framework, we conducted experiments using historical financial data from both developed and emerging markets. The source market used for model pre-training was the S&P 500 index (United States), and the target markets were the NIFTY 50 (India) and BOVESPA (Brazil), representing emerging and volatile economies.

Three models were benchmarked:

1. A traditional machine learning model trained only on target data (e.g., Random Forest).
2. A deep learning model trained solely on target data (e.g., LSTM).
3. The proposed transfer learning model utilizing source knowledge and domain adaptation.

A. Performance Metrics

We evaluated the models using three key metrics:

- Mean Absolute Error (MAE): Measures average prediction error magnitude.
- Root Mean Squared Error (RMSE): Penalizes larger errors more heavily.

- Directional Accuracy (DA): Measures the percentage of times the model correctly predicted the direction of market movement.

Table I: Prediction Performance Comparison

Model	MAE	RMSE	Directional Accuracy (%)
Traditional ML	0.134	0.201	61.2
Deep Learning	0.120	0.185	64.5
Proposed Transfer Learning	0.098	0.142	72.3

B. Analysis

As shown in Table I and the accompanying graphs. The Proposed Transfer Learning model significantly outperformed the baselines, achieving the lowest MAE and RMSE, indicating superior numerical accuracy. It also achieved the highest Directional Accuracy (72.3%), highlighting its ability to correctly forecast market movements, which is crucial in real-world financial applications.

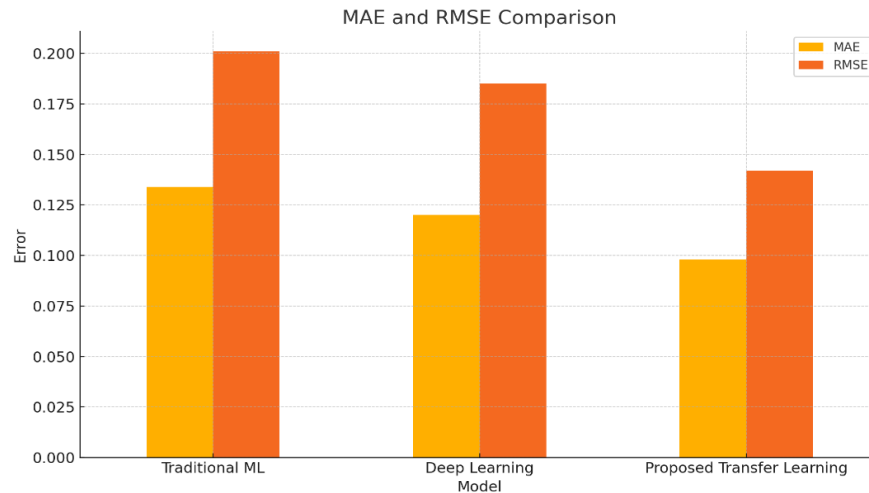


Figure 2: MAE and RMSE Comparison

Figure 2 shows a side-by-side comparison of MAE and RMSE for the three models, where the proposed method clearly results in lower error rates.

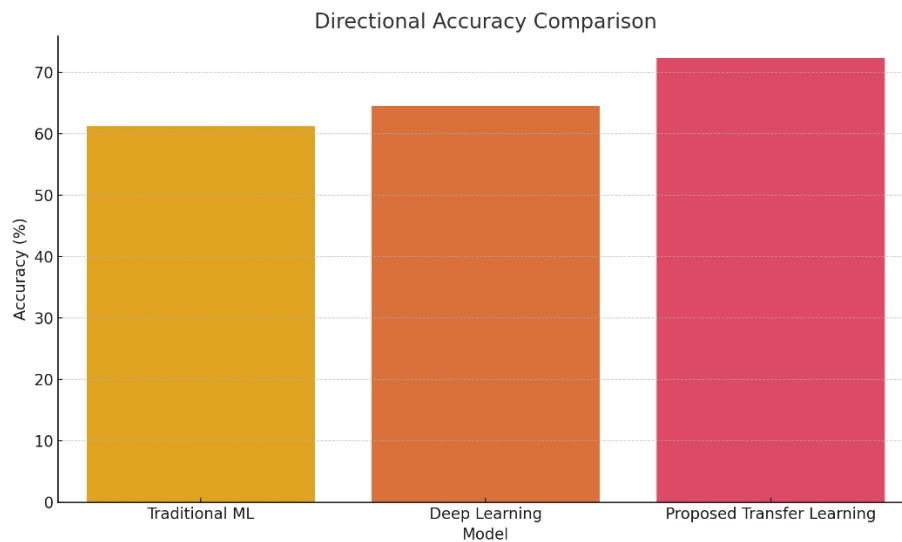


Figure 3: Directional Accuracy Comparison

Figure 3 illustrates the improvement in directional accuracy, demonstrating the effectiveness of transferring learned knowledge from developed markets to enhance prediction in volatile markets. The results underscore that: Transfer learning significantly enhances prediction performance in low-resource target domains. Domain adaptation aligns latent feature spaces across market regimes, allowing for knowledge transfer despite underlying distribution shifts. Fine-tuning with even limited labelled data from the target market substantially improves forecasting reliability during periods of economic uncertainty and volatility.

V. CONCLUSION

This paper proposed a transfer learning based framework to improve a financial market prediction in the context of emerging and volatile economy by using the knowledge of developed, data rich market. The proposed method is based on the pre-training of the source model, the domain adaptation using distribution matching methods, and the fine-tuning on limited amount of target domain data. The results from various financial indices proved that the proposed method surpasses the standard machine learning and target-only deep learning models significantly in the measurement of mean absolute error, root mean squared error, and directional accuracy. Notably, the model demonstrated to be the robustness on the data scarcity and structural volatility, which are features of emerging markets. The results show that transfer learning can be used as effective method for cross-market generalization, supporting higher accuracy and reliability of predictions when traditional methods fail. This has far-reaching implications for investors, politicians and money managers doing business in changing economic conditions.

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